

# The DRanking Protocol: Verifiable Distributed Ranking and Sybil-Bounded Admission for Structured Peer-to-Peer Networks

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## Abstract

DRanking converts locally observed peer behavior in an open structured peer-to-peer network into a globally agreed, verifiable service weight that gates admission to protocol roles. Identity creation is free, so no mechanism bounds adversarial count [1]; the protocol instead bounds adversarial weight and prices it. The instruments are: counterparty-signed service receipts in place of unsigned counters; a seed-anchored trust flow  $t_{k+1} = \alpha C_k^T t_k + (1 - \alpha)s$  obeying conservation, attenuation, and revocation laws, with a proved bound  $T_k(A) \leq \alpha g / (1 - \alpha)$  on Sybil mass; trust-weighted verifiable random sortition in place of oracle sampling; a proof-of-authority ledger trusted for ordering only, with equivocation proofs as first-class slash objects; and a circuit-friendly epoch transition folded by Nova-family incrementally verifiable computation, giving constant-cost verification to browser nodes. Rank is soulbound and slashable; reward is a separate transferable asset; the authority-to-stake transition is conditioned on a security-budget crossover.

## I. Introduction

The 2023 draft of this protocol scored behavior per identity. Identities are free; every component secured against lying nodes fell to numerous nodes. This rewrite keeps the draft's decomposition — local evidence, global aggregation, sampling, ledger, incentives — and replaces the quantity being secured:

$$\begin{aligned} \text{cost}(\text{did}) &= 0, \\ |\text{Adv} \cap V| &\leq \infty, \\ \text{secure}(| \cdot |) &= \perp \quad [1], \\ \text{secure} &:= \text{secure}(T(\cdot)), \\ T(A) &= \sum_{v \in A} t(v), \\ \text{claim} &= \text{price}(T(\text{Adv}) \geq \tfrac{1}{3}). \end{aligned}$$

Sections II–IX construct  $t$ , its evidence, its sampling, its ledger, and its economy; Section X tabulates the attack surface. Each mechanism is a term of price.

## II. Model

Definition 1 (Substrate).

$$\begin{aligned} R &= \mathbb{Z}/2^{160}\mathbb{Z} \quad [2], [3], \\ \text{did} &: K \rightarrow R, \quad \text{self-certifying}, \\ \text{link} &\subseteq V \times V, \quad \text{authenticated}, \\ \text{msg} &= \langle m, \sigma_{\text{origin}} \rangle, \\ \text{cost}(\text{did}) &= 0, \\ \text{cost}(\text{link}) &= \varepsilon > 0. \end{aligned}$$

Definition 2 (Epoch).

$$\begin{aligned} k &\in \mathbb{N}, \quad \text{epoch}(k) = [\text{fin}(\Sigma_k), \text{fin}(\Sigma_{k+1})), \\ \text{close}(k) &\equiv \text{elapsed} \geq T_{\text{ep}} \quad (\text{proposer duty}), \end{aligned}$$

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delimited by ledger finality (Definition 11); the timer  $T_{\text{ep}}$  is what enables Next in Section VIII.

Assumption 1 (Adversary).

$$\begin{aligned} |\text{Adv}| &\leq \infty, \\ \text{collude, grind, patience} &\in \text{Cap}(\text{Adv}), \\ \text{byzantine} &\in \text{Cap}(\text{Adv}), \\ \text{forge}(\sigma_h), h \notin \text{Adv} &\notin \text{Cap}(\text{Adv}), \\ \text{break}(\text{VRF}) &\notin \text{Cap}(\text{Adv}). \end{aligned}$$

Assumption 2 (Honest weight).

$$T_k(\text{Adv}) < \tfrac{1}{3} T_k(V) \quad \forall k.$$

No bound is assumed on  $|\text{Adv}|$ .

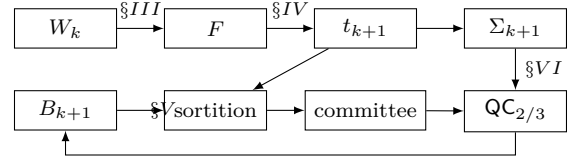


Figure 1. Epoch cycle: receipts fold into trust, trust seats the committee, the committee finalizes the ledger and threshold-signs the beacon.

## III. Service Receipts

Unsigned counters are first-person testimony; aggregated, they are hearsay. The unit of evidence is a countersigned receipt.

Definition 3 (Receipt).

$$\rho = \langle \text{srv}, a, b, k, n, \sigma_b \rangle,$$

$$\begin{aligned}
\text{srv} &\in \{\text{relay}, \text{store}, \text{probe}\}, \\
a &= \text{server}, \quad b = \text{signer/served}, \\
\text{valid}(\rho) &\equiv \text{Vrfy}(k_b, \sigma_b, \text{srv} \| a \| b \| k \| n) = \top \\
&\quad \wedge (b, k, n) \text{ fresh within } P_k,
\end{aligned}$$

freshness scoped to the epoch- $k$  input block  $P_k$  (Definition 12): the tuple carries  $k$ , so no cross-epoch nonce state exists.

Proposition 1 (Forgery surface).

$$\text{valid}(\rho) \wedge b \notin \text{Adv} \Rightarrow \rho \text{ issued by } b.$$

Proof. Immediate from  $\text{forge}(\sigma_b) \notin \text{Cap}(\text{Adv})$  (Assumption 1). Manufacturable receipts are exactly those with  $b \in \text{Adv}$  (self-dealing); these are neutralized by signer-trust weighting (Section IV).  $\square$

Definition 4 (Receipt graph).

$$\begin{aligned}
\text{val}(\rho) &= \mu_{\text{srv}(\rho)} \cdot \text{units}(\rho), \quad \mu \text{ published per kind}, \\
W_k &\in \mathbb{R}_{\geq 0}^{V \times V}, \\
W_k[b, a] &= \sum \{\text{val}(\rho) : \rho \text{ valid, sig } b, \text{srv } a, \text{ep } k\}, \\
W_k[b, \cdot] \mathbf{1} &\leq w_{\max}.
\end{aligned}$$

The cap makes countersigning a budgeted act; the protocol certifies performed service, never synthetic load (cf. Constraint 1).

#### IV. Trust Algebra

Definition 5 (Seed vector).

$$\begin{aligned}
s &\in [0, 1]^V, \quad \sum_v s_v = 1, \\
\text{supp}(s) &= S \quad (\text{authority set}).
\end{aligned}$$

Law 1 (Conservation).

$$C_k[b, a] = \frac{W_k[b, a]}{\max(w_{\max}, W_k[b, \cdot] \mathbf{1})}, \quad C_k[b, \cdot] \mathbf{1} \leq 1.$$

Law 2 (Attenuation).

$$\alpha \in (0, 1), \quad \text{flow}(\ell \text{ hops}) \leq \alpha^\ell.$$

Law 3 (Revocation).

$$\begin{aligned}
\text{slash}(v) &\Rightarrow v \notin \text{carrier} : v \text{ is excluded from} \\
&\quad \text{the vector, the columns and rows} \\
&\quad \text{of } C, \text{ and the seed vector } s, \\
t_{k+1} &= f(W_k, t_k) \quad (\text{re-derived per epoch}), \\
s \upharpoonright_{V \setminus \text{sl}} &\quad \text{un-renormalized: a slashed seed's} \\
&\quad \text{share burns; re-endorsement is a} \\
&\quad \text{threshold op (Def. 9).}
\end{aligned}$$

Without the column and seed restrictions the slash is cosmetic: receipts naming  $v$  as server, or a fixed  $s_v > 0$ , would re-grant mass to  $v$  one epoch later.

Definition 6 (Trust update).

$$t_{k+1} = \alpha C_k^\top t_k + (1 - \alpha) s. \quad (1)$$

One power-iteration step per epoch toward  $t^* = (1 - \alpha)(I - \alpha C^\top)^{-1} s$  [4], [5]; the fixpoint is approached along chain time, never computed within an epoch (cf. Definition 14).

Invariant 1 (Mass).

$$\sum_v t_0(v) = 1 \Rightarrow \sum_v t_k(v) \leq 1 \quad \forall k.$$

Proof.

$$\begin{aligned}
\sum_v t_{k+1}(v) &= \alpha \mathbf{1}^\top C_k^\top t_k + (1 - \alpha) \\
&\leq \alpha \mathbf{1}^\top t_k + (1 - \alpha) \quad (\text{Law 1}) \\
&\leq \alpha + (1 - \alpha) = 1 \quad (\text{induction}).
\end{aligned}$$

$\square$

Proposition 2 (Sybil influence bound). Partition  $V = H \uplus A$ ,  $S \subseteq H$ . Let

$$g_k = \sum_{b \in H, a \in A} C_k[b, a] t_k(b), \quad g = \sup_k g_k.$$

Then

$$T_k(A) = \sum_{v \in A} t_k(v) \leq \frac{\alpha g}{1 - \alpha} \quad \forall k.$$

Proof.  $S \cap A = \emptyset$  kills the seed term over  $A$ . Splitting (1) by origin region and applying Law 1 to internal circulation,

$$T_{k+1}(A) \leq \alpha(g_k + T_k(A)).$$

From  $T_0(A) = 0$ ,

$$T_k(A) \leq \alpha g \sum_{i \geq 0} \alpha^i = \frac{\alpha g}{1 - \alpha}.$$

$\square$

$T(A)$  is linear in honest endorsement granted ( $g$ ), independent of  $|A|$ . Defense concentrates on  $g$ : endorsement spends budget (Law 1), and detection claws it back (Law 3).

Proposition 3 (Farming price). Instantiating Proposition 2 at  $A = \{v\}$ : sustaining weight  $w$  requires per-epoch honest endorsement

$$g_v \geq \frac{1 - \alpha}{\alpha} w \quad \text{for every epoch the weight is held,}$$

so

$$\text{farm}(w, \ell \text{ epochs}) \geq \ell \cdot \text{cost}\left(\frac{1 - \alpha}{\alpha} w\right) + \text{Pr}[\text{detect}] \cdot u(w),$$

where  $u(w)$  is the holder's valuation of the forfeited stock (same cost units as the first term); the second term by Law 3: one accepted equivocation forfeits the stock. This is the **farm** function of Proposition 14, made explicit.

Proof.  $T_k(\{v\}) \leq \alpha g_v / (1 - \alpha)$ ; rearrange, multiply by holding duration, add the slash expectation.  $\square$

Proposition 4 (Geometric convergence). For stationary  $C_k = C$ , the map  $\Phi(x) = \alpha C^\top x + (1 - \alpha)s$  satisfies

$$\|\Phi(x) - \Phi(y)\|_1 \leq \alpha \|x - y\|_1,$$

hence  $t^*$  is the unique fixpoint and

$$\|t_k - t^*\|_1 \leq \alpha^k \|t_0 - t^*\|_1.$$

Proof.

$$\begin{aligned}
\|C^\top(x - y)\|_1 &= \sum_a \left| \sum_b C[b, a](x_b - y_b) \right| \\
&\leq \sum_b \left( \sum_a C[b, a] \right) |x_b - y_b| \leq \|x - y\|_1
\end{aligned}$$

by Law 1; Banach on the affine contraction. Under drifting  $C_k$  the iterate tracks the moving fixpoint with lag  $O(\alpha^j)$  per  $j$  epochs of stability.  $\square$

Definition 7 (Soulbound rank).

$$\begin{aligned} \text{rank}_k(v) &= t_k(v), \\ \text{transfer} &\notin \text{Ops}(\text{rank}), \\ \text{rank} &\xrightarrow{\text{slash}} 0, \\ \text{rank} &\xrightarrow{\text{epoch}} \text{re-derived (1)}. \end{aligned}$$

(Cf. Proposition 14.)

Definition 8 (Dilution schedule).

$$\begin{aligned} \text{vol}_k &= \sum_{i \leq k} \mathbf{1}^\top W_i \mathbf{1}, \\ 1 - \alpha_k &= D(\text{vol}_k), \quad D \text{ non-increasing, published,} \\ \bar{\alpha} &= \sup_k \alpha_k < 1, \\ t_0 &= s \quad (\text{cold start}). \end{aligned}$$

Propositions 2–4 hold under the time-varying schedule with  $\alpha := \bar{\alpha}$ .

Definition 9 (Threshold authority).

$$\begin{aligned} \text{Ops}(S) &= \{\text{withdraw, rotate,} \\ &\quad \text{endorse}_0, \text{reshare}\}, \\ o \in \text{Ops}(S) &\Rightarrow \text{QC}_\theta(S)(o), \quad \theta \geq \frac{2}{3}, \\ |\text{comp}| < \theta|S| &\Rightarrow \text{Cap}(\text{Adv}) \cap \text{Ops} = \emptyset, \\ \text{keys} &: \text{genesis DKG; rotate and} \\ &\quad \text{beacon-group change trigger} \\ &\quad \text{proactive reshare,} \\ &\quad \text{preserving gpk.} \end{aligned}$$

## V. Sortition

Proposition 5 (Statistical undetectability). Model  $\text{did} = \text{h}(\text{pk})$ ,  $\text{h}$  a random oracle. For the adversarial placement strategy “sample fresh keys,” honest and adversarial positions are identically distributed on  $R$ :

$$\Delta_{\text{TV}}(\text{pos}(H \uplus A), \text{pos}(H')) = 0 \quad \forall |A|,$$

hence every position-based test (entropy, Kolmogorov–Smirnov) has distinguishing advantage 0 against Sybil density.

Proof.  $\text{h}(\cdot)$  uniform on  $R$  for every key, honest or adversarial; a union of i.i.d. uniforms is distribution-invariant in its partition.  $\square$

Placement tests detect biased sampling; the threat is cheap identity. Validation must be cryptographic and economic, and a global random seed is a bootstrap circularity (a randomness beacon presupposes the committee being selected). Both are removed by sortition [6], [7].

Definition 10 (Beacon).

$$\begin{aligned} G_0 &= S \quad (\text{genesis DKG}), \\ G_k &= \text{vset}(\Sigma_k), \\ \text{TSig}_G &: \theta\text{-of-}|G|, \text{ unique [8]:} \\ &\quad |\{\sigma : \text{TVrfy}(\text{gpk}, k, \sigma)\}| = 1 \quad \forall k, \\ B_k &= \text{h}(\text{TSig}_{G_{k-1}}(k)), \\ \text{gpk} &\in \Sigma \quad (\text{invariant under reshare}). \end{aligned}$$

The group is inductive, which breaks the bootstrap circularity in both phases: the already-seated group  $G_{k-1}$  signs the beacon that seats epoch  $k$ ; proactive resharing hands the same  $\text{gpk}$  to each new  $\text{vset}$ , so verification never changes. Remark: all  $B_k$  are determined at key generation; unpredictability holds only below  $\theta$  shares (as in drand-style beacons) — a  $\theta$ -quorum can precompute every future beacon, which Definition 9 already assumes it must not form adversarially.

Proposition 6 (Beacon unbiasedability).

$$B_k = \text{fn}(\text{gpk}, k) \wedge \text{predict}(B_k) \Rightarrow \theta \text{ shares},$$

i.e. no participant choice influences  $B_k$ . A header-hash beacon  $\text{h}(\text{fin}(\Sigma_k))$  is excluded: proposers grind  $B_k$  through content selection.

Proof. Uniqueness leaves a single valid  $\sigma$  per  $k$ : the signing protocol has no free variable to grind. Unpredictability below  $\theta$  shares is the threshold-signature security game.  $\square$

Algorithm 1 Trust-weighted sortition (node  $v$ , epoch  $k$ , role  $r$ )

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- 1:  $B_k \leftarrow \text{h}(\text{TSig}_{G_{k-1}}(k))$   $\triangleright$  Definition 10
  - 2:  $(y, \pi) \leftarrow \text{VRF}_{\text{sk}_v}(B_k \| r)$   $\triangleright \text{sk}_v$  = the DID key
  - 3:  $\text{seats}_v \leftarrow \text{InvCDF}_{\text{Bin}(\hat{t}_k(v), \tau/\hat{T}_k)}(y/2^{|y|})$
  - 4:  $v$  holds  $\text{seats}_v$  seats for  $r$ ; publishes  $\pi$
- 

Weight is atomized: node  $v$ ’s fixed-point weight  $\hat{t}_k(v)$  (Definition 16) is  $\hat{t}_k(v)$  unit atoms,  $\hat{T}_k = \sum_v \hat{t}_k(v)$ ; each atom is selected independently with probability  $\tau/\hat{T}_k$ , realized per node through the inverse binomial CDF on the VRF output [6]. Atomization removes both hidden hypotheses of naive indicator sortition: expected seats are exactly proportional to weight with no per-node cap ( $\mathbb{E}[\text{seats}_v] = \tau \hat{t}_k(v)/\hat{T}_k$ , whales simply hold several seats), and normalization by  $\hat{T}_k$  makes the fraction well-defined even when total mass is below one (Invariant 1). The DID key doubles as the VRF key: it is committed by  $v$ ’s first appearance in the ledger, hence fixed before any later beacon exists (Proposition 7).

Proposition 7 (Grinding closure).

$$\begin{aligned} \text{sk}_v \text{ fixed} &\prec B_k \text{ exists,} \\ B_k \text{ unbiasedable} &\quad (\text{Proposition 6}), \quad \Rightarrow \text{grind} = \perp. \end{aligned}$$

Proof. Both VRF inputs are committed before either party can condition on the output; unpredictability from Assumption 1 and Proposition 6.  $\square$

Proposition 8 (Committee safety). Atoms are independent indicators, so with  $\pi_A = \hat{T}(A)/\hat{T} < \frac{1}{3}$  (Assumption 2, which is fractional),  $\mathbb{E}[X] = \tau$  and  $\mathbb{E}[X_A] = \tau \pi_A$ . By Chernoff–Hoeffding over the atom indicators,

$$\Pr[X_A \geq \frac{\tau}{3}] \leq \exp\left(-\tau D\left(\frac{1}{3} \parallel \pi_A\right)\right),$$

with  $D$  the binary relative entropy: committee dishonesty decays exponentially in committee size. Sortition is the

transport of Assumption 2 from weight to seats; no reported quantity exists for a coalition to steer, so the draft's median game is vacuous by construction.

## VI. Admission Ledger

Definition 11 (Ledger).

$$\begin{aligned}\Sigma_k &= \langle \text{rt}(t_k), \text{rt}(V_k), \text{rt}(\text{sl}_k), \text{rt}(\text{bal}_k), \text{gpk}, \text{ver} \rangle, \\ \text{fin}(\Sigma_k) &\equiv \text{QC}_{2/3}(\text{vset}(\Sigma_{k-1})) \quad [9], [10], \\ \text{content}_k &= \{\text{rt}(X_k)\}, \quad \text{data}(X_k) \in \text{Overlay}.\end{aligned}$$

Definition 12 (Input block).

$$\begin{aligned}X_k &= \langle P_k, E_k, \text{TX}_k, \text{fees}_k \rangle, \\ P_k &= \text{receipt multiset}, \\ E_k &= \text{equivocation proofs}, \\ \text{TX}_k &= \text{transfer/bond/unbond ops}, \\ W_k &= \text{Agg}(P_k) : \text{deterministic fold} \text{ ---} \\ &\quad \text{dedup } (b, k, n) \text{ within } P_k, \text{ rows} \\ &\quad \text{clipped to } w_{\max} \text{ in hash order}.\end{aligned}$$

$X_k$  is assembled by the slot proposer from submissions during  $\text{epoch}(k)$ ;  $\text{Agg}$  is a pure function of the multiset, so  $W_k$  is canonical given  $X_k$ . Proposer discretion over admission to  $X_k$  is exactly the censorship power bounded by Obligation 2.

Obligation 1 (Minimal trust).

$$\begin{aligned}\forall o \in \text{content} : \text{Verify}(o) &\perp \text{ honest}(S), \\ \text{Cap}(\text{quorum}) &= \{\text{order}, \text{censor}\}, \\ \text{forgeable}(\text{quorum}) \ni o &\Rightarrow \text{design defect}.\end{aligned}$$

Definition 13 (Equivocation proof).

$$e = \langle m_1, \sigma_v(m_1), m_2, \sigma_v(m_2) \rangle,$$

$$\text{slot}(m_1) = \text{slot}(m_2) \wedge m_1 \neq m_2 \Rightarrow \text{slash}(v),$$

verification assumption-free;  $e$  is a first-class ledger object, submittable by anyone.

Obligation 2 (Force inclusion). Submissions are acknowledged per slot proposer  $p$ :

$$\text{ack}_p(x) \wedge \text{proposes}(p, [k, k + \Delta]) \wedge x \notin X_{[k, k + \Delta]} \Rightarrow \text{slash}(p),$$

with evidence  $\langle \text{ack}_p(x), p$ 's signed block  $\rangle$  — both signed by  $p$ , hence self-certifying and attributable: the slash lands on the proposer who acknowledged and then withheld, never on a bystander quorum member. Censorship, the residual validator power, is thereby bounded by latency  $\Delta$  times the number of colluding proposers a submitter must route around.

Equivocation defeats patience: months of farmed standing forfeit on one contradiction, so  $\mathbb{E}[\text{farm}] \downarrow$  with detection probability.

## VII. Folded State

Browser wasm nodes can neither replay history nor audit a validator signature; the chain must be succinct [11].

Definition 14 (Epoch transition).

$F(\Sigma_k, X_k) = \Sigma_{k+1}$ ,  $\text{rt}(X_k)$  public input,  $X_k$  witness, typed against Definition 12: the body consumes the opened block (receipts, proofs, transactions, fees), never bare roots.

Algorithm 2  $F$  — pure, deterministic, per epoch

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1:  $\text{sl}_{k+1} \leftarrow \text{sl}_k \cup \{\text{verify}(e) : e \in E_k\}$ 
2:  $U \leftarrow V \setminus \text{sl}_{k+1}$ ;  $t' \leftarrow t_k \upharpoonright_U$ ;  $s' \leftarrow s \upharpoonright_U$ ;  $\tilde{C}_k \leftarrow C_k \upharpoonright_{U \times U}$  ▷ Law 3
3:  $t_{k+1} \leftarrow \alpha_k \tilde{C}_k^T t' + (1 - \alpha_k) s'$  ▷ Eq. (1), Def. 8
4:  $\text{bal}' \leftarrow \text{apply}(\text{TX}_k, \text{bal}_k)$  ▷ transfers, bond/unbond;
    $\text{bal} \geq \text{amt}$ 
5:  $\text{bal}_{k+1} \leftarrow \text{bal}' + \text{issue}(W_k, t_k) - \text{burn}(\text{fees}_k)$ 
6:  $V_{k+1} \leftarrow \text{vset}(\Sigma_{k+1})$  ▷ Obligation 8
7: return  $\langle \text{rt}(t_{k+1}), \text{rt}(V_{k+1}), \text{rt}(\text{sl}_{k+1}), \text{rt}(\text{bal}_{k+1}), \text{gpk}, \text{ver} \rangle$ 

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$$\begin{aligned}\text{cost}(F) &= O(\text{nnz}(W_k) + |\text{TX}_k|) \\ &\quad \text{adds/mults over committed data}.\end{aligned}$$

Line 2 restricts vector, matrix (both dimensions), and seeds: rows die with  $t'(b) = 0$ , but columns and  $s$  must be cut explicitly or slashed nodes re-accrue mass (Law 3). Line 3 is why the fixpoint is unrolled: one sparse matrix-vector product arithmetizes trivially; an eigenvector solve does not. Line 4 is the transfer semantics of the two-asset economy (Definition 18): **apply** is sum-preserving, and bond/unbond move balance into and out of **Bond** with an unbonding delay of  $\Delta_{\text{ub}}$  epochs (slash exposure persists through the delay).

Definition 15 (Folding). Nova-family IVC [12]:

$$\begin{aligned}\pi_k &\vdash \Sigma_0 \xrightarrow{F^k} \Sigma_k, \\ |\pi_k| &= O(1), \quad \text{Verify}(\pi_k) = O(1), \\ \text{sync} &= \text{Verify}(\pi_k) \wedge \text{QC}_{2/3}(\Sigma_k), \\ \text{prove} &\parallel \text{consensus} \quad (\text{asynchronous, trails head}).\end{aligned}$$

Proposition 9 (Linearity).

$$\text{fin}(\Sigma_k) \text{ instant (BFT)} \Rightarrow \text{chain} \cong (\Sigma_0, \Sigma_1, \dots) \text{ linear},$$

hence IVC applies with no re-folding branch; probabilistic-finality chains admit no such reduction.

Proof. Quorum intersection: two finalized blocks at one height require  $\geq \frac{1}{3}$  of the finality committee to equivocate (Definition 13) — contradicting Definition 9 in the authority phase, and Proposition 8 with Assumption 2 in the staked phase.  $\square$

Obligation 3 (Circuit discipline).

$$\begin{aligned}\forall \text{ extension } F' : \\ F' \text{ pure} \wedge F' \in \text{CheapArith}(\text{committed}).\end{aligned}$$

The ledger is an admission ledger, not a contract platform.

Obligation 4 (Verification boundary).

in-fold :  $F$  (state arithmetic over  $X_k$ ),  
 out-of-fold :  $\text{Vrfy}(\rho)$ ,  $\text{VRF } \pi$ ,  $\text{TSig}$ ,  
                   checked by validators pre-QC,  
 $\text{QC}(\Sigma' \parallel \text{rt}(X)) \Rightarrow$  attests validity,  
 invalid  $\rho \in X_k \Rightarrow$  fraud proof from  $X_k$  data  
                    $\Rightarrow$  quorum slash.

Elliptic-curve verifications are not **CheapArith** (Obligation 3); the fold proves the arithmetic transition, the quorum attests input validity, and anyone holding  $\text{data}(X_k)$  can re-check — so forged inputs are not impossible but detectable and slashable, preserving Obligation 1 through accountability rather than circuit cost. In the staked phase the sortition proofs seating  $V_{k+1}$  live on the same side: verified out-of-fold, consumed by line 6 of Algorithm 2 as committed input.

Definition 16 (Fixed-point carrier). Circuits have no reals. With precision  $\phi$ :

$$\begin{aligned} \hat{t} &= \lfloor t \cdot 2^\phi \rfloor \in \mathbb{Z}_{\geq 0}, \\ \hat{x} \cdot \hat{y} &\mapsto \lfloor \hat{x} \hat{y} / 2^\phi \rfloor, \\ x/y &\mapsto (q, r) : x = qy + r, \\ &0 \leq r < y \quad (\text{witnessed, ranged}). \end{aligned}$$

Proposition 10 (Rounding soundness). Floor rounding is one-sided ( $\hat{t}_k \leq t_k 2^\phi$ : weight is never overcounted, so safety bounds survive verbatim), and the accumulated error is damped by the contraction of Proposition 4:

$$e_{k+1} \leq \bar{\alpha} e_k + n 2^{-\phi} \Rightarrow \sup_k e_k \leq \frac{n 2^{-\phi}}{1 - \bar{\alpha}},$$

bounded for all time; precision is a parameter, not a drift.

Proof. Each of the  $n$  components loses at most  $2^{-\phi}$  per epoch to truncation; the prior error enters only through  $\bar{\alpha} C^\top$ , which is  $\ell_1$ -nonexpansive scaled by  $\bar{\alpha}$ ; geometric sum.  $\square$

Proposition 11 (Carrier boundedness).

$$|\text{carrier}(\hat{t}_k)| \leq 2^\phi \quad \forall k,$$

independent of  $|V|$ : truncation is garbage collection.

Proof. Invariant 1 in fixed point gives  $\sum_v \hat{t}_k(v) \leq 2^\phi$  atoms; every carried  $v$  holds  $\hat{t}_k(v) \geq 1$  atom, and a component truncated to 0 leaves the carrier (its Merkle leaf is pruned). Without truncation the carrier grows without bound — geometric decay never reaches zero — so state boundedness is a free consequence of Definition 16, not an extra mechanism.  $\square$

Obligation 5 (Upgrade versioning).

$$\begin{aligned} F \mapsto F' &\Rightarrow \text{vk} \mapsto \text{vk}', \\ \text{ver} &\in \Sigma, \\ \text{bind}(\text{vk}, \text{vk}') &\text{ at committed epoch;} \end{aligned}$$

shipped at genesis, not retrofitted.

Folding provides validity only:  $\text{canonicity} \in \text{consensus}$ ,  $\text{availability} \in \text{overlay}$ .

Definition 17 (Commitment granularity).

$$\begin{aligned} \Gamma &\in \{\text{full}, \text{agg}, \text{blind}\}, \\ \text{leak}(\text{full}) &= \{(b, a, W[b, a])\} \quad (\text{the traffic graph}), \\ \text{leak}(\text{agg}) &= \{(v, \text{deg}^\pm(v))\}, \\ \text{leak}(\text{blind}) &= \emptyset, \\ \Gamma \neq \text{full} &\Rightarrow F \text{ consumes edge data as private} \\ &\quad \text{witness (blind: ZK openings),} \\ \text{replay}(\Gamma) &= \top \iff \Gamma = \text{full}. \end{aligned}$$

$\Gamma$  trades public replayability against surveillance of the service graph; the choice is a standing design decision, and  $\Gamma = \text{blind}$  prices its privacy in circuit cost (Obligation 3).

## VIII. State Relation

The ledger state machine in TLA form [13], the model-checking surface for finite instances [14]:

$$\begin{aligned} \text{Spec} &\triangleq \text{Init} \wedge \Box[\text{Next}]_\Sigma \wedge \text{WF}_\Sigma(\text{Next}), \\ \text{Init} &\triangleq \Sigma = \langle \text{rt}(s), \text{rt}(S), \emptyset, \text{rt}(\mathbf{0}), \text{gpk}_0, \text{ver}_0 \rangle, \\ \text{Next} &\triangleq \exists X : \text{QC}_{2/3}(\Sigma' \parallel \text{rt}(X)) \wedge \Sigma' = F(\Sigma, X), \end{aligned}$$

with  $\text{gpk}_0$  from the genesis DKG (Definition 9).

Invariant 2 (Slash monotone).

$$\text{sl} \subseteq \text{sl}', \quad v \in \text{sl} \Rightarrow t(v) = 0.$$

Proof. Lines 1–3 of Algorithm 2:  $\text{sl}$  only grows by union, and the update runs entirely over  $U = V \setminus \text{sl}_{k+1}$  — vector  $t'$ , both dimensions of  $\hat{C}_k$ , and seeds  $s'$  are restricted, so no term of line 3 can assign mass to a slashed index. (Restricting the vector alone would not suffice: an unrestricted column of  $C_k$ , or a fixed  $s_v > 0$ , re-grants mass one epoch later — hence Law 3 demands all three cuts.)  $\square$

Invariant 3 (Version monotone).

$$\text{ver}' \geq \text{ver},$$

by Obligation 5:  $\text{vk}$  rebinding occurs only at committed epoch boundaries, forward.

Invariant 4 (Unique chain).

$$\text{fin}(\Sigma_k) \wedge \text{fin}(\Sigma'_k) \Rightarrow \Sigma_k = \Sigma'_k,$$

by Proposition 9 (quorum intersection under Assumption 2).

Proposition 12 (Progress). Under partial synchrony (after global stabilization time) with a live honest quorum —  $\theta|S|$  members in the authority phase (Definition 9), honest committee majority via Proposition 8 in the staked phase —

$$\Box \Diamond \text{Next-enabled} \wedge \text{WF}_\Sigma(\text{Next}) \Rightarrow \forall k : \Diamond \text{fin}(\Sigma_k).$$

Proof.  $W$  may be empty, so **Next** is enabled whenever a quorum is live; BFT view-change liveness [9] delivers  $\text{QC}_{2/3}$  after stabilization; weak fairness converts enabledness into occurrence.  $\square$

Obligation 6 (Checkable instances). Every implementation change to  $F$  ships with a finite-instance model ( $|V| \leq n_0$ , bounded epochs) checking Invariants 1, 2–4 before merge.

## IX. Economics

Definition 18 (Two assets).

$$\begin{aligned} \text{rank} &: \text{soulbound (Def. 7),} \\ &\quad \text{gates admission/sortition,} \\ \text{reward} &: \text{transferable via TX (Alg. 2, 1.4),} \\ \text{issue}_k(v) &\propto (W_k^\top t_k)(v). \end{aligned}$$

Proposition 13 (Countersigning game). For a service-consuming node  $b$ :

$$\begin{aligned} \text{cost}(\text{sign}) &= 0 \quad (\text{outflow spends } b\text{'s row,} \\ &\quad \text{never } t(b)), \\ u_b(\text{withhold}) &= \text{choked: service needs receipts,} \\ u_b(\text{sign}) &= \text{continued service,} \end{aligned}$$

so truthful countersigning is weakly dominant under link-layer tit-for-tat (the eDonkey/BitTorrent heritage [15], [16]). Selling receipts is not free-riding but attack-edge formation: the seller's flow enters  $g$  of Proposition 2, is volume-capped by  $w_{\max}$ , and is priced by Proposition 3.

Proposition 14 (Payable rank defeats cost anchoring). Let rank be transferable at price  $p$ . Then

$$\begin{aligned} \text{cost}_{\text{Adv}}(m) &= \min(\text{farm}(m), p \cdot m), \\ u_{\text{exit}}(\text{hold}) = 0 \wedge p > 0 &\Rightarrow \text{exiting nodes sell,} \\ \text{supply} > 0 &\Rightarrow p \leq \inf_v \text{farm}_v(m)/m, \end{aligned}$$

so acquisition cost decouples from behavior. Non-transferability removes the second branch of the min, restoring  $\text{farm}(m) = \text{service} + \text{time} + \text{slash risk}$  as the sole path.  $\square$

Invariant 5 (Balance conservation).

$$\mathbf{1}^\top \text{bal}_{k+1} = \mathbf{1}^\top \text{bal}_k + \text{issue}_k - \text{burn}_k,$$

by lines 4–5 of Algorithm 2:  $\text{apply}(\text{TX}_k, \cdot)$  is sum-preserving (transfers permute balance, bond/unbond move it between  $\text{bal}$  and  $\text{Bond}$  within the same total), so supply changes only through published issuance and burn.

Definition 19 (Fee loop).

$$\begin{aligned} \text{Sinks} &= \{\text{store}_{>\rho}, \text{QoS}, \text{namespace}\}, \\ \text{fees} &\rightarrow \text{burn} \cup \text{issue pool}, \\ \text{anchor}_{\text{PoS}} &\leq f(\text{demand}(\text{Sinks})). \end{aligned}$$

Obligation 7 (Security-budget crossover).

$\text{activate}(\text{PoS}) \iff \text{Bond} \geq \beta \cdot \widehat{\text{payoff}}(\text{attack}), \quad \beta > 1,$   
conditioned, never scheduled; pre-crossover, PoA finality overlays PoS production, weighted down by  $D(\text{vol}_k)$  (Definition 8).

Obligation 8 (Validator-set abstraction).

$$\begin{aligned} \text{vset} &: \Sigma \rightarrow 2^V, \\ \text{vset}_{\text{PoA}} &= \text{const } S, \\ \text{vset}_{\text{PoS}} &= \text{sortition}(\text{Bond}) \\ &\quad (\text{seat proofs out-of-fold, Obl. 4}); \end{aligned}$$

the transition changes one pure function, not the engine.

Constraint 1 (Rejected anchors).

$$\begin{aligned} \text{PoW} &: \text{cost}(51\%) = \text{rent}(\text{hash}) \cdot O(\text{hours}) \Rightarrow \perp, \\ \text{PoT} &: W_{\text{synth}}[b, a], \quad a, b \in \text{Adv} \not\subseteq W_{\text{real}} \\ &\Rightarrow \text{bandwidth burned, } \Delta T(\text{Adv}) \approx 0 \Rightarrow \perp. \end{aligned}$$

Receipted real service with trust weighting is proof-of-traffic minus the synthetic load.

## X. Security Analysis

Table I  
Threat matrix

| Attack          | Bound                                                   | Residual                  |
|-----------------|---------------------------------------------------------|---------------------------|
| Sybil flood     | Prop. 2                                                 | $\alpha g / (1 - \alpha)$ |
| Self-dealing    | Prop. 1 + signer-trust weight; cap (Def. 4)             | zero-trust circulation    |
| Patient farming | Laws 1–3; total slash (Def. 13)                         | undetected deviation      |
| Whitewashing    | fresh DID $\Rightarrow t = 0$ ; slow earn, instant loss | re-entry at zero          |
| Grind/eclipse   | Prop. 7                                                 | —                         |
| Censorship      | Obl. 2                                                  | latency $\Delta$          |
| Seed compromise | threshold $S$ ; revocation; Def. 8                      | pre-dilution window       |
| Rank market     | Prop. 14                                                | out-of-band key sale      |

$$\begin{aligned} \text{sale}(\text{sk}_v) &\Rightarrow \text{buyer inherits slash exposure,} \\ \text{PoA} &= \text{trust root until Obl. 7.} \end{aligned}$$

Table II  
Protocol parameters

| Sym.                 | Role                    | Constraint                                              |
|----------------------|-------------------------|---------------------------------------------------------|
| $\alpha_k$           | damping / seed weight   | $\bar{\alpha} = \sup_k \alpha_k < 1$ ; Def. 8           |
| $w_{\max}$           | receipt issuance cap    | bounds per-identity sale volume                         |
| $\tau$               | expected committee size | error $e^{-\tau D(1/3 \  T(A))}$ , Prop. 8              |
| $\theta$             | authority threshold     | $\geq 2/3$ ; Defs. 9, 10                                |
| $\Delta$             | inclusion timeout       | censorship latency, Obl. 2                              |
| $T_{\text{ep}}$      | epoch timer             | enables Next, Def. 2                                    |
| $\Delta_{\text{ub}}$ | unbonding delay         | slash exposure persists, Def. 14                        |
| $\mu$                | service metering        | published per kind, Def. 4                              |
| $\phi$               | fixed-point precision   | error $\leq n2^{-\phi} / (1 - \bar{\alpha})$ , Prop. 10 |
| $D$                  | dilution curve          | non-increasing, published, Def. 8                       |
| $\beta$              | crossover margin        | $> 1$ , Obl. 7                                          |

## XI. Related Work

Local scoring descends from eDonkey queues [15] and BitTorrent choking [16]; propagation from EigenTrust and personalized PageRank [4], [5], here unrolled along epochs over signed evidence; the attack-edge bound pattern from

SybilGuard/SybilLimit/SybilRank [17]–[19], instantiated over a service graph. Brahms [20] addresses adversarial sampling that the draft’s tests could not (Proposition 5); DRanking sidesteps via Algorand-style sortition [6], [7] weighted by trust. S/Kademlia [21] prices identifier creation (complementary one-time cost). Finality follows PBFT lineage [9], [10]; succinctness follows Mina [11] via Nova folding [12]; impossibility framing is Douceur’s [1]; the external-cost anchor observation is Nakamoto’s [22].

## XII. Conclusion

$$\text{safe} \iff \text{cost}(T(\text{Adv}) \geq \tfrac{1}{3}) > \widehat{\text{payoff}}(\text{attack}),$$

and every mechanism above — receipts, the trust laws, sortition, the ledger, folding, the two-asset economy — is a term on the left-hand side.

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